Symbol	Parameter	Conditions	Min	Typ	Max	Unit
C_{ADC}	Internal sample and hold capacitor	-	-	3	-	pF
$t_{ADCVREG\_STUP}$	ADC LDO startup time	-	-	-	10	µs
t_{STAB}	ADC power-up time	LDO already started	1	-	-	conversion cycle
t_{OFF_CAL}	Offset calibration time	-	85			$1/f_{ADC}$
t_{LATR}	Trigger conversion latency for regular and injected channels	Trigger from an asynchronous clock	3	-	4	
		Trigger from a synchronous clock	3			
t_S	Sampling time	-	2.5	-	288.5	
t_{CONV}	Total conversion time (including sampling time)	N-bits resolution	t_S + 1.5 + N			
$I_{DDA(ADC)}$	ADC consumption on V_{DD}, V_{DDA} and V_{REF}	f_S = 2.25 MSPS	-	200	-	µA

1. *High temperature generate leakage current on ADC inputs. This current create voltage drop through Rain directly affecting ADC accuracy (worst case when $V_{AIN} = V_{DDA}$). To limit this effect to a tolerance of 2LSBs, you must respect Rain max tables.*

Table 55. 12-bit ADC accuracy

ADC accuracy values are measured after internal calibration. Resolution = 12 bits, no oversampling.
Evaluated by characterization on LQFP100. Packages without a VREF- pad can have degraded specifications. Not tested in production.

Symbol	Parameter	Min	Typ	Max	Unit
ET	Total unadjusted error	-	4	5.7	LSB
EO	Offset error	-	±2.5	±5	
EG	Gain error	-	2.6	6	
ED	Differential linearity error	-1	-	1.3	
EL	Integral linearity error	-3	±2.5	3	
ENOB	Effective number of bits	-	10.7	-	bits
SINAD	Signal-to-noise and distortion ratio	-	66	-	
SNR	Signal-to-noise ratio	-	68	-	dB
THD	Total harmonic distortion	-	-70	-	